

■ 555 Timer IC

Astable Oscillator

Detailed Lesson Handout • Grade 4 • Electronics Basics

Subject	Basic Electronics
Grade	4 (Ages 9–10)
Duration	45–60 minutes
Prepared by	Planrs Academy

■ Learning Objectives

By the end of this lesson, students will be able to:

- ✓ Explain what an Integrated Circuit (IC) is using everyday analogies.
- ✓ Describe the 555 Timer IC – its appearance, cost, and core purpose.
- ✓ Define **astable mode** and explain why it makes an LED blink automatically.
- ✓ Identify all 8 pins of the 555 Timer IC and state the function of each.
- ✓ Explain the roles of resistors (RA, RB) and a capacitor (C) in controlling blink speed.
- ✓ Connect real-world applications (traffic signals, Diwali lights) to the 555 Timer circuit.
- ✓ Answer quiz questions and complete hands-on homework activities.

■ Section 1 – What Is an Integrated Circuit (IC)?

An **Integrated Circuit (IC)** is a tiny electronic component that packs many smaller parts – resistors, capacitors, transistors – onto a single chip of silicon. Think of it like a school **tiffin box (dabba)**: the box itself is small, but inside it holds many separate food items neatly arranged. An IC works the same way!

Tiffin Box (Dabba)	Integrated Circuit (IC)
The box itself	The chip package (plastic/ceramic body)
Food compartments	Individual electronic components inside
Holds rice, curry, roti separately	Holds resistors, transistors, capacitors
Saves space vs. carrying separate containers	Saves space vs. using discrete parts
Costs a few rupees	Costs ₹5–₹10 (very affordable!)

*Key vocabulary: **IC** = Integrated Circuit; **Silicon** = the material chips are made from; **Component** = a single electronic part.*

■ Section 2 – Meet the 555 Timer IC

The **555 Timer IC** is one of the most popular and widely used ICs in the world. It was invented in **1972** by Hans Camenzind and has been used in billions of devices ever since. Despite being over 50 years old, it is still manufactured today because it is so versatile and affordable.

■ Key Facts About the 555 Timer IC

- Full name: NE555 / LM555 Timer Integrated Circuit
- Package: 8-pin DIP (Dual In-line Package) – looks like a small black rectangle with legs
- Supply Voltage: Works on 4.5 V to 15 V (a 9V battery is perfect!)
- Cost in India: Approximately ₹5 – ₹10 (very budget-friendly)
- Modes: Can work in 3 modes – Astable, Monostable, Bistable
- Output Current: Can drive an LED directly (up to ~200 mA)
- Still produced by companies like Texas Instruments, STMicroelectronics

■ Section 3 – Astable Mode (The Dholak Player)

What does 'Astable' mean?

The word **astable** comes from the Greek prefix *a-* meaning 'not' and *stable* meaning 'steady'. So astable literally means **not stable** - the circuit never settles into one fixed state. Instead, it keeps switching between HIGH (ON) and LOW (OFF) continuously, forever, as long as power is supplied.

The Dholak Analogy ■

Imagine a dholak (drum) player at a wedding. The player beats the drum in a steady, repeating rhythm – BOOM-BOOM-BOOM – never stopping. The 555 Timer in astable mode does exactly the same thing with electricity: it sends out a steady ON-OFF-ON-OFF rhythm that we call a **square wave**. This rhythm makes the LED blink!

The square wave output of the 555 Timer (visualised as time slots):

HIGH (LED ON)	LOW (LED OFF)	HIGH (LED ON)	LOW (LED OFF)
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The width of each slot depends on resistor and capacitor values – this controls how fast the LED blinks.

■ Section 4 – The 8 Pins of the 555 Timer IC

The 555 Timer IC has **8 pins** – tiny metal legs that let us connect it to other components on a breadboard or PCB. Each pin has a specific job. Understanding the pins is the key to building any 555 circuit!

Pin	Name	Function
1	GND	Ground – negative terminal of power supply
2	Trigger	Starts the output HIGH when voltage drops below $V_{cc}/3$
3	Output	The blinking signal comes out here → connects to LED
4	Reset	Resets timer; keep connected to V_{cc} to keep it running
5	Control	Voltage control – connect $0.01\ \mu\text{F}$ capacitor to ground
6	Threshold	Stops output HIGH when voltage exceeds $2/3\ V_{cc}$
7	Discharge	Discharges timing capacitor – connects to resistors
8	VCC	Positive power supply (+4.5 V to +15 V)

Memory Tip: The dot or notch on one end of the chip marks Pin 1. Pins 1–4 run down the left side; pins 5–8 run up the right side.

■ Section 5 – Resistors, Capacitors & Blink Speed

5.1 What is a Resistor?

A **resistor** limits the flow of electric current – just like a **speed breaker** on a road slows down cars. The higher the resistance (measured in **Ohms** – Ω), the less current flows through. In the 555 astable circuit we use two resistors:

Resistor	Role in Circuit	Typical Value	Analogy
RA	Controls charge time (LED ON period)	10 k Ω	Speed breaker on the way UP
RB	Controls discharge time (LED OFF period)	47 k Ω	Speed breaker on the way DOWN

5.2 What is a Capacitor?

A **capacitor** stores electric charge and then releases it – just like a **clay pot (matka)** fills with water and then pours it out. In the 555 circuit, the capacitor (C) charges through RA + RB and discharges through RB. This charge-discharge cycle creates the timing for the blink.

State	What the Capacitor Does	LED State
Charging ↑	Fills with electricity through RA + RB	ON ■

Fully charged	Voltage reaches 2/3 VCC → threshold triggers	Switching...
Discharging ↓	Empties through RB only	OFF ■
Fully discharged	Voltage drops to 1/3 VCC → trigger fires again	Switching...

Golden Rule of Blink Speed: Bigger RA, RB or C values → **Slower** blinking. Smaller RA, RB or C values → **Faster** blinking.

5.3 Timing Formulae (for curious learners!)

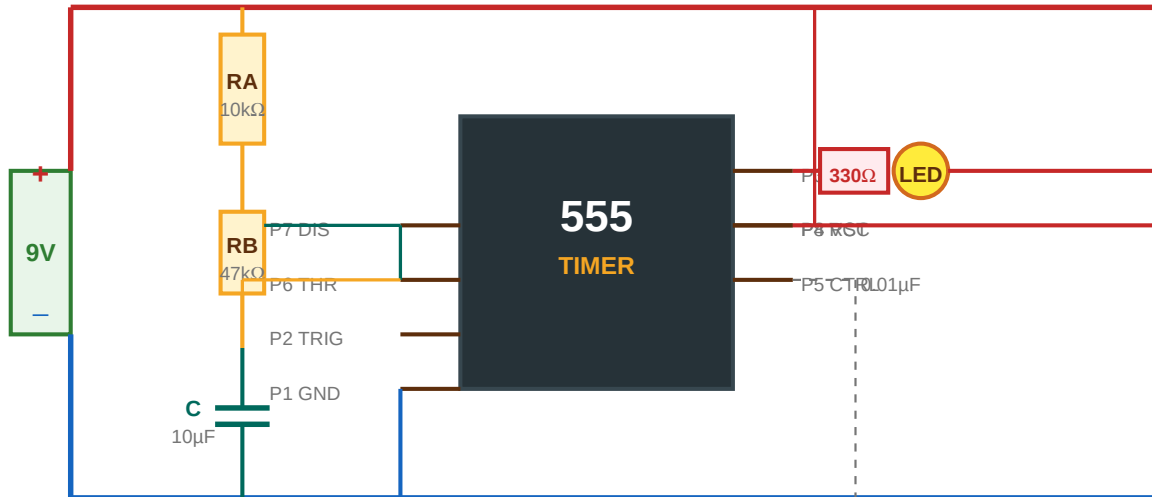
The exact timing is calculated using these simple formulae:

Formula	What it Calculates
$t_{\text{HIGH}} = 0.693 \times (R_A + R_B) \times C$	Time the LED stays ON
$t_{\text{LOW}} = 0.693 \times R_B \times C$	Time the LED stays OFF
$T = t_{\text{HIGH}} + t_{\text{LOW}}$	Total time for one complete blink cycle
$f = 1 / T$	Frequency – how many blinks per second (Hz)

Example: $R_A = 10 \text{ k}\Omega$, $R_B = 47 \text{ k}\Omega$, $C = 10 \text{ }\mu\text{F}$ gives approximately $t_{\text{HIGH}} \approx 0.39 \text{ s}$, $t_{\text{LOW}} \approx 0.33 \text{ s}$, $f \approx 1.4 \text{ Hz}$ (about 1–2 blinks per second).

■ Section 6 – Circuit Diagram

The diagram below shows how the 555 Timer IC is wired in astable mode to make an LED blink. Study it carefully and trace the path of electricity!



Astable 555 Oscillator – simplified schematic (not to scale)

Component	Value / Type	Purpose in Circuit
555 Timer IC	NE555 or LM555	Generates the automatic ON/OFF switching
Resistor RA	10 kΩ (Brown-Black-Orange)	Controls LED ON time
Resistor RB	47 kΩ (Yellow-Violet-Orange)	Controls LED OFF time
Capacitor C	10 μF electrolytic	Sets the blink period
Capacitor C2	0.01 μF ceramic	Stabilises control voltage (Pin 5)
LED	Red / Green / Blue	Visible output – blinks on and off
Resistor R_LED	330 Ω	Protects LED from too much current
Battery	9 V	Power supply for the circuit

■ Section 7 – Real-World Applications in India

The 555 Timer IC is everywhere around us! Learning to use it means you already understand how many everyday devices work. Here are some examples from daily life in India:

■ Railway & Traffic Signals

Red/amber/green lights blink in a timed rhythm – controlled by oscillator circuits similar to the 555 Timer. Railway crossing warning lights blink automatically with no human pressing buttons.

■ Bicycle Tail Lights

Safety lights on bicycles flash rapidly so drivers can see cyclists at night. 555-based flasher circuits are very common in affordable Indian bicycle lights.

■ Diwali Decorative Lights

Twinkling LED strings used during Diwali and other festivals use 555 Timer circuits to create the alternating blink and chase light effects.

■ Doorbell Chimes

Many simple Indian doorbells use a 555 Timer in astable mode to produce a tone through a small buzzer for a set duration when the button is pressed.

■ Electronic Timers

Kitchen timers, school bell systems, and irrigation controllers all rely on 555-based timing circuits to count seconds and minutes reliably.

■ Alarm Systems

Simple burglar alarm sirens and smoke alarm beepers use 555 Timers to generate the repeating high-pitched beep that alerts people.

⇒ Section 8 – Quick Quiz

Q1. What does an IC do?

- a) Packs many electronic parts into one tiny chip
- b) Makes food
- c) Plays music

Answer: a) Packs many electronic parts into one tiny chip

Q2. Astable mode means the LED ...

- a) Stays ON always
- b) Blinks ON and OFF forever automatically
- c) Never turns ON

Answer: b) Blinks ON and OFF forever automatically

Q3. What is a resistor like?

- a) A speed breaker that slows electricity
- b) A water pot that stores electricity
- c) A drum that beats electricity

Answer: a) A speed breaker that slows electricity

Q4. Which pin of the 555 Timer outputs the blinking signal?

- a) Pin 1 (GND)
- b) Pin 3 (Output)
- c) Pin 8 (VCC)

Answer: b) Pin 3 (Output)

Q5. If you use a BIGGER capacitor, what happens to the blink speed?

- a) Blinks faster
- b) Blinks slower
- c) Does not change

Answer: b) Blinks slower

Q6. What is the cost of a 555 Timer IC in India (approximately)?

- a) ₹500
- b) ₹50
- c) ₹5–₹10

Answer: c) ₹5–₹10

■ Section 9 – Key Vocabulary Glossary

Integrated Circuit (IC)	A tiny chip containing many electronic components packed together.
555 Timer IC	A popular timer IC used to generate timing pulses or square waves.
Astable	Not stable; continuously switches between HIGH and LOW states.
Oscillator	A circuit that produces a repeating signal (like a blinking LED or a buzzer tone).
Square Wave	An electrical signal that switches between fully ON and fully OFF at regular intervals.
Resistor	A component that limits the flow of electric current. Unit: Ohms (Ω).
Capacitor	A component that stores and releases electric charge. Unit: Farads (F).
Frequency	How many times per second the LED blinks. Unit: Hertz (Hz).
VCC	Positive power supply terminal of the IC (also written V+).
GND (Ground)	Negative power supply terminal – the common reference point.
Pin	A metal leg of the IC that connects it to the rest of the circuit.
Threshold	A voltage level that triggers a change in the IC's output.
Breadboard	A reusable plastic board with holes for prototyping circuits without soldering.
LED	Light-Emitting Diode – a component that lights up when current flows through it.
Duty Cycle	The percentage of time the output is HIGH compared to the total cycle time.

■ Section 10 – Homework & Activities

Complete these activities before the next session:

⇒ Activity 1 – Draw Your IC

Draw the 555 Timer IC from memory. Label all 8 pins with their numbers and names (GND, Trigger, Output, Reset, Control, Threshold, Discharge, VCC). Add colour – use red for power pins and blue for ground!

■ Activity 2 – Blinker Hunt

Look around your home or neighbourhood and find 3 real examples of blinking lights. Write them down and try to guess if a 555 Timer (or something similar) is inside. Examples: indicator lights, fan speed indicators, emergency lights.

■ Activity 3 – Calculate a Blink!

Use the formula $f = 1.44 / ((RA + 2 \times RB) \times C)$. Try: $RA = 10 \text{ k}\Omega$, $RB = 47 \text{ k}\Omega$, $C = 10 \text{ }\mu\text{F}$. What is the frequency? (Hint: convert $\text{k}\Omega$ to Ω by multiplying by 1,000; convert μF to F by dividing by 1,000,000.)

■ Activity 4 – Teach a Family Member

Explain what a 555 Timer IC is to a family member using the tiffin box analogy. Then explain what astable mode means using the dholak player analogy. Ask them to give you a star rating out of 5 for your explanation!

■ Activity 5 – Explore (Optional)

Ask a parent to search 'Tinkercad 555 Timer Astable Circuit' online. You can simulate a blinking LED circuit in the browser for free – no components needed!

■ Lesson Summary – What We Learned

- ✓ IC = tiffin box with many electronic parts packed inside.
- ✓ 555 Timer IC = small, cheap (~₹5–10), powerful timer chip with 8 pins.
- ✓ Astable mode = dholak rhythm that never stops → LED blinks automatically.
- ✓ Pin 1 = GND | Pin 3 = Output (LED connects here) | Pin 8 = VCC.
- ✓ Resistors (RA , RB) = speed breakers controlling charge/discharge timing.
- ✓ Capacitor (C) = matka filling and emptying to set the blink period.
- ✓ Bigger R or C = slower blink. Smaller R or C = faster blink.

✓ 555 Timer is used in traffic signals, Diwali lights, doorbells, alarms.